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Questions

Q1.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

A gas molecule of mass *m* travelling at speed *v* strikes the side of a gas cylinder at an angle *θ*. The collision takes time *t*.



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What is the average force acting on the side of the gas cylinder?

☐ **A** $\frac{mv}{t}$

☐ **B** $\frac{2mv}{t}$

☐ **C** $\frac{mv\sin\theta}{t}$

☐ **D** $\frac{2mv\sin\theta}{t}$

(Total for question = 1 mark)

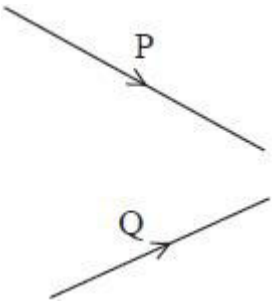
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Q2.

The diagram shows the momenta P and Q of two bodies about to collide.

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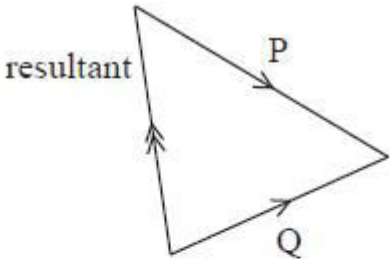
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When the bodies collide they merge and move on as a single body.

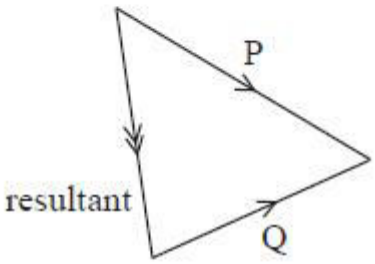
Which of the following diagrams shows the resultant momentum?

☐ **A**

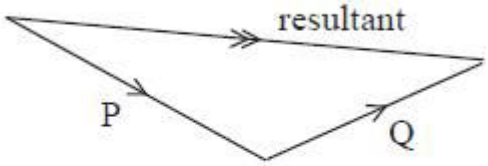


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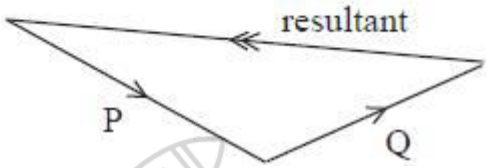
☐ **B**



☐ **C**



☐ **D**



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(Total for question = 1 mark)

Q3.

The photograph shows a jumping toy.

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The head is pushed down onto the base, compressing a spring. When released the spring expands and the toy jumps into the air.

A student investigated the toy.

The student placed the toy on a balance to measure the force required to compress the spring. The force was 14 N when the spring was fully compressed by 1.7 cm.

When the toy was launched it jumped to a height of 1.5 m.

(a) Show that the maximum energy stored in the compressed spring was about 0.1 J.

(2)

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(b) Initially the spring was fully compressed. As the toy was released, the head accelerated upwards on its own until the spring was uncompressed. The head then pulled the base upwards and they moved together at the same speed.



spring fully compressed



head accelerates upwards



head pulls base

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The student used the elastic potential energy stored in the spring to determine the maximum speed of the head.
The student used the gravitational potential energy gained by the whole toy to determine its initial upwards speed.
Determine whether the toy obeys the law of conservation of momentum using values of speed calculated in this way.
mass of head = 6.4 g
mass of whole toy = 7.2 g
maximum height of jump = 1.5 m

(5)

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(c) Determine whether kinetic energy was conserved as the head began to pull the base upwards.

(2)

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(Total for question = 9 marks)

Q4.

A ball of mass *m* travelling with velocity *v* strikes a wall at right angles. It bounces off the wall in the opposite direction at the same speed.

Which of the following is the impulse on the wall?

☒ **A** mv

☒ **B** $2mv$

☒ **C** $-mv$

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☒ **D** $-2mv$

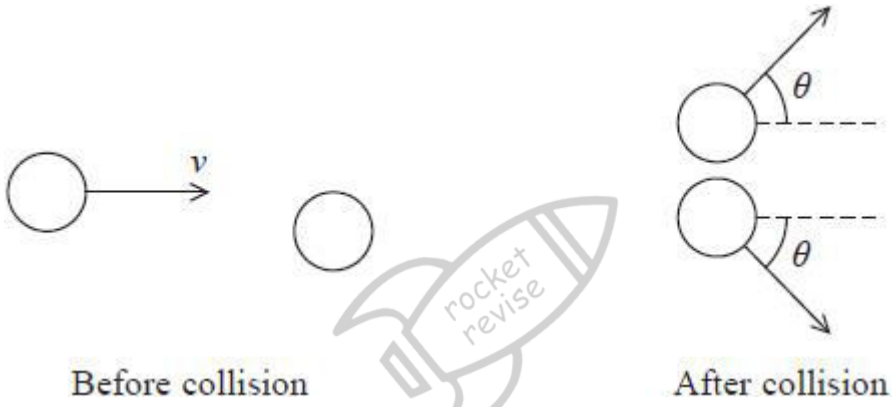
(Total for question = 1 mark)

Q5.

A sphere travelling at speed v collides elastically with an identical sphere which is at rest.

After the collision, both spheres move off at an angle θ to the direction of travel of the first sphere, as shown. The spheres have the same speed as each other.

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What is the speed of the spheres after the collision?

☒ **A** v

☒ **B** $\frac{v}{\sqrt{2}}$

☒ **C** $\frac{v}{2}$

☒ **D** $\frac{v}{4}$

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(Total for question = 1 mark)

Q6.

Some asteroids pass very close to the Earth. Scientists are planning methods to deflect asteroids, to prevent them hitting the Earth.

One method would involve colliding a spacecraft into the surface of the asteroid, to change the path and speed of the asteroid. The spacecraft would remain joined to the asteroid after the collision.

(a) This collision method is modelled for a spacecraft travelling in a direction at 90° to the path of the

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asteroid.

Sketch a labelled vector diagram to show the momenta of the bodies before and after the collision.

(2)

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(b) Show that the momentum of the spacecraft is about 10^7 N s.

mass of spacecraft = 920 kg
speed of spacecraft = 12 000 m s^{-1}

(2)

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(c) Show that this collision method causes the asteroid to change its direction through an angle of about 10^{-7} radian.

momentum of asteroid = 7.6×10^{13} N s

(2)

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(d) After the collision, the asteroid and spacecraft remain joined and move together.

Calculate the component of their velocity at 90° to the original path of the asteroid after the collision.
mass of asteroid = 2.8×10^9 kg

(2)

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Component of velocity = ...

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(e) Another method would involve attaching a rocket motor to the asteroid and using the motor to apply a force to the asteroid.

In this method the force is applied at 90° to the path of the asteroid.
Deduce whether this would produce a change in momentum as great as the change produced by the collision method.
force exerted by rocket motor = 5.1×10^6 N
time for which rocket motor applies force = 6 minutes

(2)

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(Total for question = 10 marks)

Q7.

A student carried out experiments on momentum using two table hockey pucks, as shown.



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The pucks each contain a small fan, so that they glide across the table on a cushion of air.
The mass of the pucks can be varied by attaching small masses.

In each experiment, the student pushed one puck towards a stationary puck.

(a) In one experiment, the first puck reached a speed of 0.35 m s^{-1} after being pushed for a time of 0.28 s.

Calculate, using the idea of impulse, the average force used to accelerate the first puck.
mass of puck = 110 g

(3)

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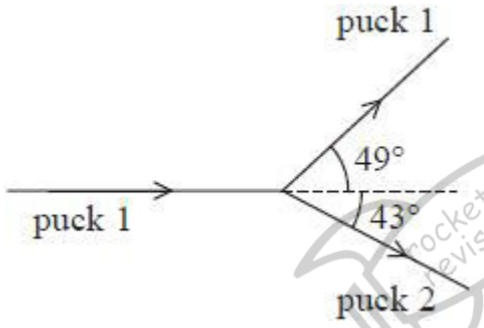
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Average force = ...

(b) In another experiment, the first puck was pushed towards the stationary puck with a speed of 0.41 m s^{-1}.

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The paths of the pucks before and after the collision are shown. The paths are labelled puck 1 and puck 2.



(i) Calculate the speed of puck 2 after the collision.
mass of puck 1 = 110 g
mass of puck 2 = 130 g
speed of puck 1 after collision = 0.28 m s^{-1}

(4)

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Speed of puck 2 = ...

(ii) Deduce whether this was an elastic collision.

(3)

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(c) Explain the assumption made when applying the principle of conservation of momentum to collisions.

(2)

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(Total for question = 12 marks)

Q8.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A particle of mass m has kinetic energy E_k and momentum p. A second particle of mass $2m$ has kinetic energy $2E_k$.

Both particles are non-relativistic.

Which of the following is equal to the momentum of the second particle?

☐ **A** $\sqrt{2}p$

☐ **B** p

☐ **C** $2p$

☐ **D** $4p$

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(Total for question = 1 mark)

Q9.

A particle of mass m has momentum p and kinetic energy E_k.
A second particle of mass $m/2$ has momentum $2p$.

What is the kinetic energy of the second particle?

☐ **A** $E_k/8$

☐ **B** $E_k/2$

☐ **C** $2\,E_k$

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☒ **D** 8 E_k

(Total for question = 1 mark)

Q10.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which row of the table identifies what happens to momentum and kinetic energy in an elastic collision?

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		Momentum	Kinetic energy
☒	**A**	conserved	conserved
☒	**B**	conserved	not conserved
☒	**C**	not conserved	conserved
☒	**D**	not conserved	not conserved

(Total for question = 1 mark)

Q11.

Select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question refers to a collision between two trolleys.

The diagram shows the momentum of two trolleys, X and Y, before a collision. The mass of each trolley is 0.25 kg.

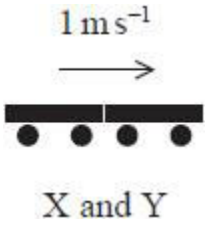
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The two trolleys join together after the collision and move on with a velocity of 1 m s^{-1} .

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Which of the following is the kinetic energy of trolley X before the collision?

- ☒ **A** 0.5 J
- ☒ **B** 1 J
- ☒ **C** 2 J
- ☒ **D** 4 J

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(Total for question = 1 mark)

Q12.

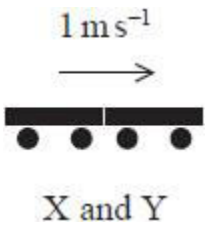
Select one answer from A to D and put a cross in the box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question refers to a collision between two trolleys.

The diagram shows the momentum of two trolleys, X and Y, before a collision. The mass of each trolley is 0.25 kg.



The two trolleys join together after the collision and move on with a velocity of 1 m s^{-1} .



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Which row of the table is correct for this collision?

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	Momentum	Elastic or inelastic collision
☒ A	conserved	elastic
☒ B	conserved	inelastic
☒ C	not conserved	elastic
☒ D	not conserved	inelastic

(Total for question = 1 mark)

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Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

An object of mass *m* has momentum *p* and kinetic energy *E*.

Which of the following is the mass of an object with momentum 2*p* and kinetic energy 4*E*?

☒ **A** $\frac{1}{2}m$

☒ **B** m

☒ **C** $2m$

☒ **D** $4m$

(Total for question = 1 mark)

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Q14.

A helicopter can hover in a fixed position as shown.

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(Source: Ascent Xmedia/Getty Images)

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The helicopter blades move air vertically downwards.

Explain how this enables the helicopter to maintain a constant height above the ground.

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(Total for question = 4 marks)

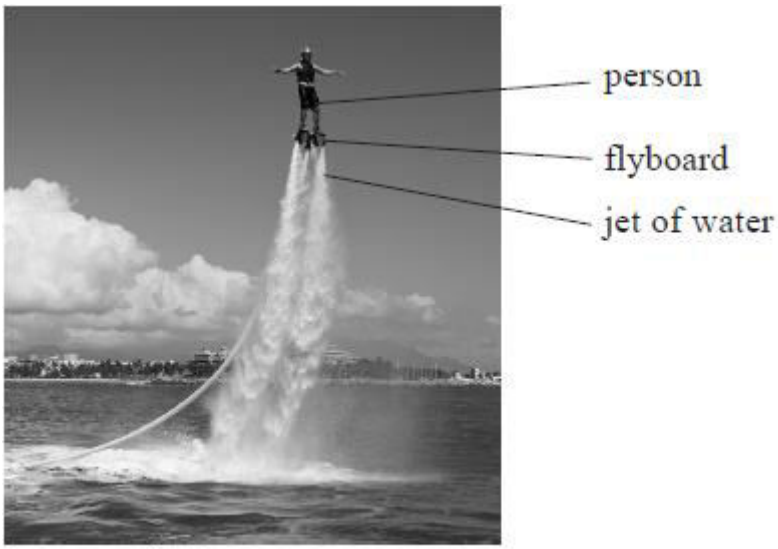
Q15.

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A flyboard enables a person to hover at a constant height above the sea as shown. Water is constantly pumped up to the flyboard in a thick pipe. A jet of water is then forced downwards, causing an upwards force on the flyboard.

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(Source: © Justin Lewis/Getty Images)

Calculate the velocity of the jet of water as it leaves the flyboard. Assume the water has negligible velocity before it leaves the flyboard.

mass of person and flyboard equipment = 175 kg
mass flow rate of water = 114 kg s^{-1}

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Velocity of water jet = ...

(Total for question = 3 marks)

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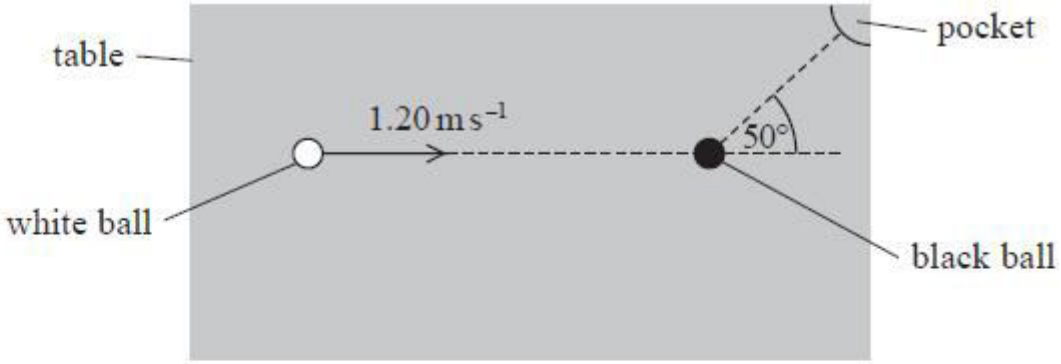
Q16.

In a game of snooker, a white ball and a black ball of equal mass are on a horizontal table. A player hits the white ball which then moves with a velocity of 1.20 m s^{-1} before colliding with the black ball. The player hopes that the collision will knock the black ball towards the pocket at the corner of the table as shown.

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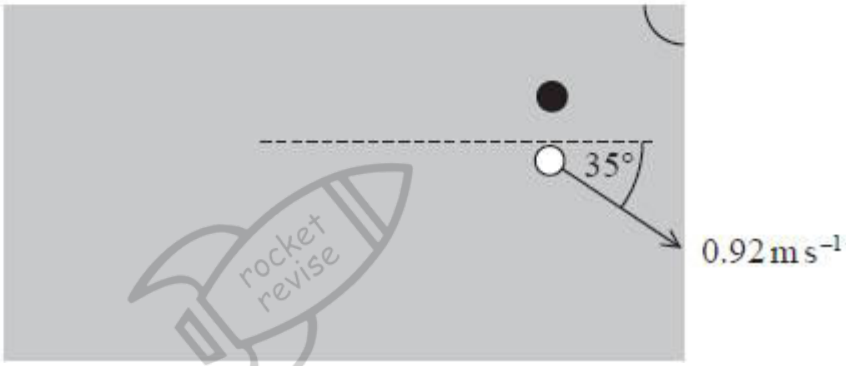
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Plan view



After the collision, the velocity of the white ball was at 35° to its original path, as shown below.

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(a) This collision was inelastic.

State what is meant by inelastic.

(1)

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(b) For this situation, a scaled vector diagram showing the velocities of the balls can be used to demonstrate the law of conservation of momentum.

Explain why.

(2)

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(c) Deduce whether the black ball moves towards the pocket. You should use a scaled vector diagram.

(5)

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(Total for question = 8 marks)

Q17.

A student used a spreadsheet to model the conservation of momentum.

(a) State the principle of conservation of momentum.

(2)

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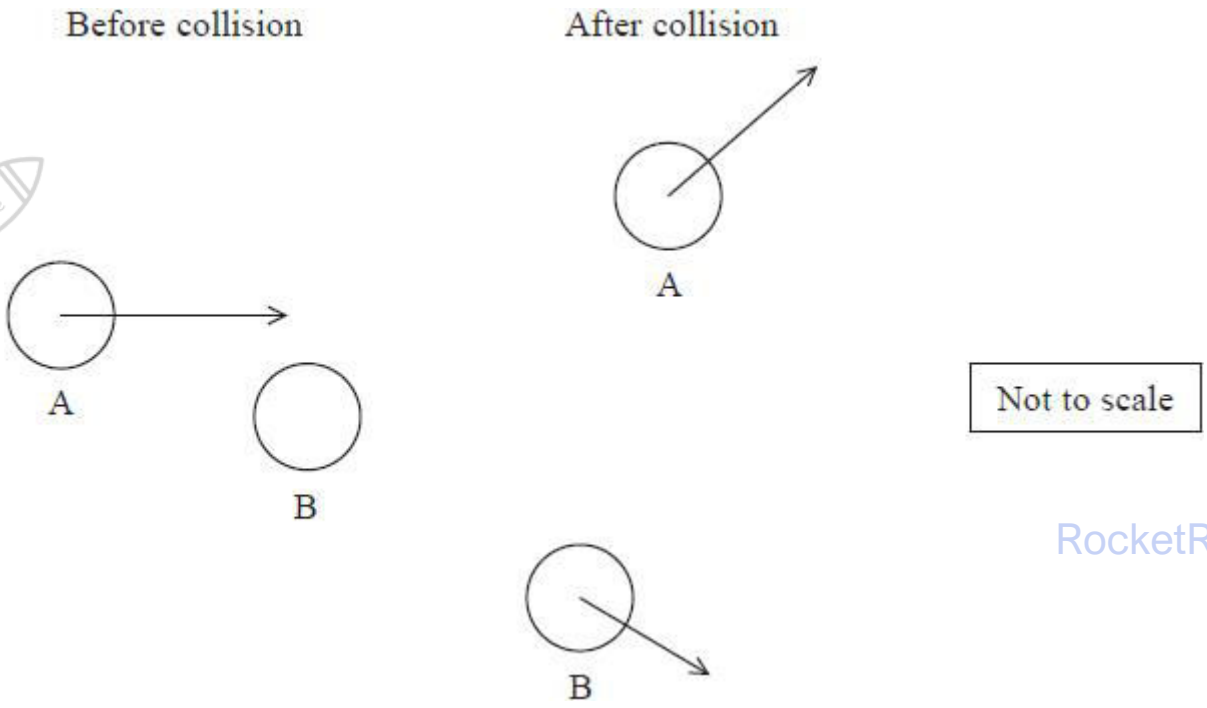
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(b) The spreadsheet modelled an elastic collision between object A and object B.

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The spreadsheet shows the initial values and final values when the student tested the model.

	A	B	C	D	E
1		Initial values		Final values	
2		object A	object B	object A	object B
3	mass / kg	0.85	1.70	0.85	1.70
4	magnitude of velocity / m s^{-1}	1.30	0.00	0.98	0.54
5	angle of velocity to x direction / °	0.0	0.0	54.5	–48.0
6					

(i) Deduce whether the values show an elastic collision.

(3)

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(ii) Deduce whether the values show the conservation of momentum.

(5)

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(Total for question = 10 marks)

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Q18.

An electron has kinetic energy 7.2×10^{-16} J.

Calculate the de Broglie wavelength of the electron.

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Wavelength = ...

(Total for question = 3 marks)

Q19.

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Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A car travels along a straight horizontal road. A drag force of 150 N acts for 15 s causing the speed of the car to decrease.

Which of the following is the change in momentum of the car due to the drag force?

- ☐ **A** 10 kg m s^{-1}
- ☐ **B** 1125 kg m s^{-1}
- ☐ **C** 2250 kg m s^{-1}
- ☐ **D** 16 875 kg m s^{-1}

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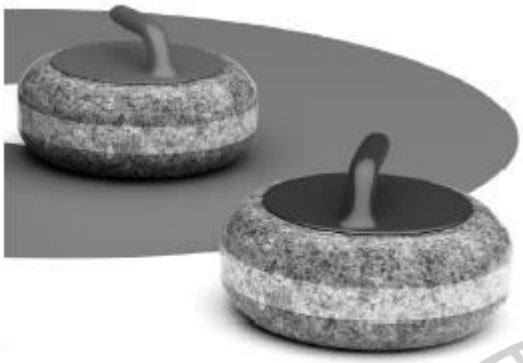
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(Total for question = 1 mark)

Q20.

Curling is a sport played on ice. Players slide stones made of granite along the ice.

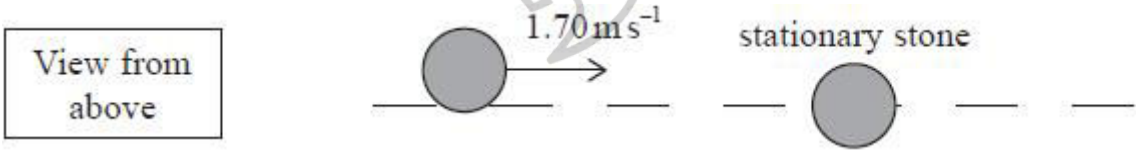
The photograph shows two of the stones. Each stone has a mass of 19.1 kg.



(Source: © MileA/Getty Images)

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A player slides a stone at a velocity of 1.70 m s^{-1} towards a stationary stone, as shown below.



The stones collide. After the collision, both stones move off as shown below.



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(a) Show that *v* is about 1.3 m s^{-1}.

(4)

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(b) Deduce whether the collision was elastic. Your answer should include a calculation.

(3)

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(Total for question = 7 marks)

Q21.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which of the following gives the base units of impulse?

- ☐ **A** kg m s^{-1}
- ☐ **B** kg m s^{-2}
- ☐ **C** N m
- ☐ **D** N s

(Total for question = 1 mark)

Q22.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A ball travelling at speed *u* towards a wall collides with the wall at right angles. The ball bounces off the wall at speed *v* in the opposite direction. The time for the collision is *t*.

Which of the following expressions gives the average force on the wall during the collision?

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☒ **A** $\frac{m(u+v)}{t}$

☒ **B** $\frac{m(u+v)}{2t}$

☒ **C** $\frac{m(u-v)}{t}$

☒ **D** $\frac{m(u-v)}{2t}$

(Total for question = 1 mark)

RocketRevise

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A non-relativistic particle of mass m has momentum p and kinetic energy E_k .

A second non-relativistic particle of mass $\frac{m}{2}$ has momentum $2p$.

Which of the following expressions gives the kinetic energy of the second particle?

☒ **A** $\frac{E_k}{2}$

☒ **B** E_k

☒ **C** $2E_k$

☒ **D** $8E_k$

(Total for question = 1 mark)

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Q24.

The diagram shows the track of a $\sum^+$ particle in a bubble chamber. The particle decays at point X.

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There is a magnetic field of magnetic flux density *B* perpendicular to the velocity of the particle.

(a) The $\sum^+$ particle is following a circular path of radius 1.99 m.

Calculate the value of *B*.
momentum of $\sum^+$ particle = 3.67×10^{-19} kg m s^{-1}

(2)

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B = ...

(b) The equation shows the decay of the $\sum^+$ particle.

$$\sum^+ \rightarrow \pi^+ + n$$

(i) The $\sum^+$ particle is made of two up quarks and one strange quark.
State what type of particle $\sum^+$ is.

(1)

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(ii) Deduce whether the conservation of charge and the conservation of lepton number apply to this decay.

(4)

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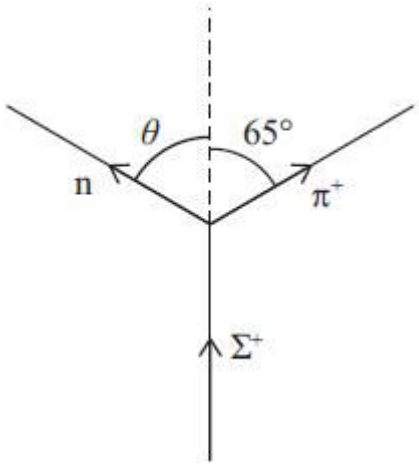
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(iii) The diagram shows the directions of the three particles involved in the decay.

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Not to scale



Determine the magnitude and direction of the momentum of the neutron.
momentum of $\sum^+$ particle = 3.67×10^{-19} kg m s^{-1}
magnitude of momentum of π^+ = 1.07×10^{-19} kg m s^{-1}

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(6)

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Magnitude of momentum = ...

Direction of momentum = ...

(iv) The diagram shows the track of the $\sum^+$ particle in the bubble chamber.

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Add to the diagram to show all tracks visible in the bubble chamber after the decay at X.
You should label your diagram.

(4)

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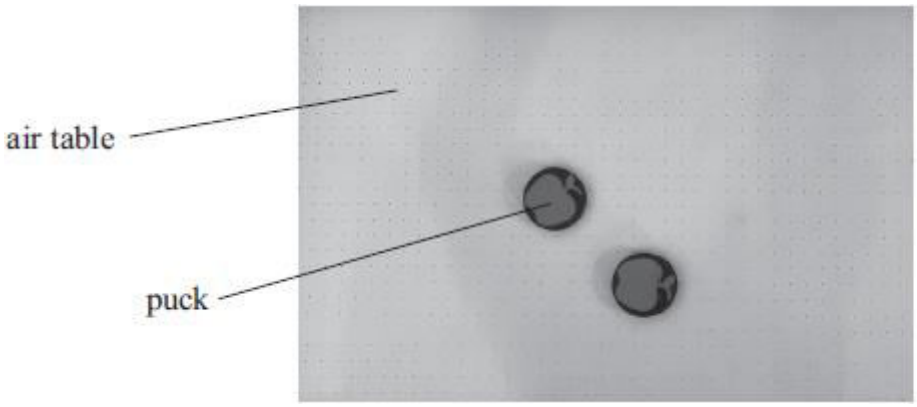
(Total for question = 17 marks)

Q25.

An air table has a surface with many small holes. Air is blown through the holes.

Plastic pucks can move freely over the table on a cushion of air.

The photograph shows the surface of an air table with two pucks on it.



View from above

Some students used the air table to investigate conservation of momentum.

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(a) Explain how using the air table ensured that momentum was conserved in collisions between the pucks.

(2)

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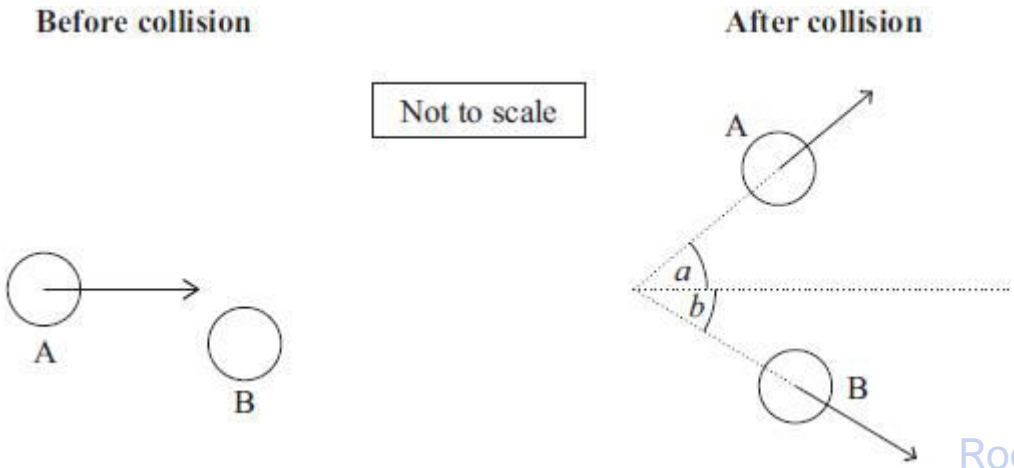
(b) The students observed several collisions between two identical pucks, A and B.

One puck was stationary before each collision.

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They noticed that after each collision the two pucks seemed to follow paths at 90° to each other. The diagram shows one of the collisions.



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The students recorded the following data for this collision.

initial momentum of A	$0.046\,\text{kg m s}^{-1}$
angle a	33°
final momentum of A	$0.039\,\text{kg m s}^{-1}$

(i) Deduce whether the angle between the paths of the pucks after the collision was 90°. You should use the principle of conservation of momentum.

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(ii) Deduce whether the collision was elastic.
mass of each puck = 0.110 kg

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(Total for question = 12 marks)

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Q26.

Many sub-atomic particles are unstable and have only been detected in high energy particle collision experiments.

(a) Kaons are short-lived sub-atomic particles. They are also known as K-mesons and were the first strange particles
to be discovered.

(i) The following table lists some quarks and their charge.

Quark	Charge / e
Up (u)	$+\frac{2}{3}$
Down (d)	$-\frac{1}{3}$
Strange (s)	$-\frac{1}{3}$

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Explain the quark structure of a K^- meson.

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(ii) The mass of a K^- meson is 494 MeV / c^2.
Calculate the mass of a K^- meson in kg.

(3)

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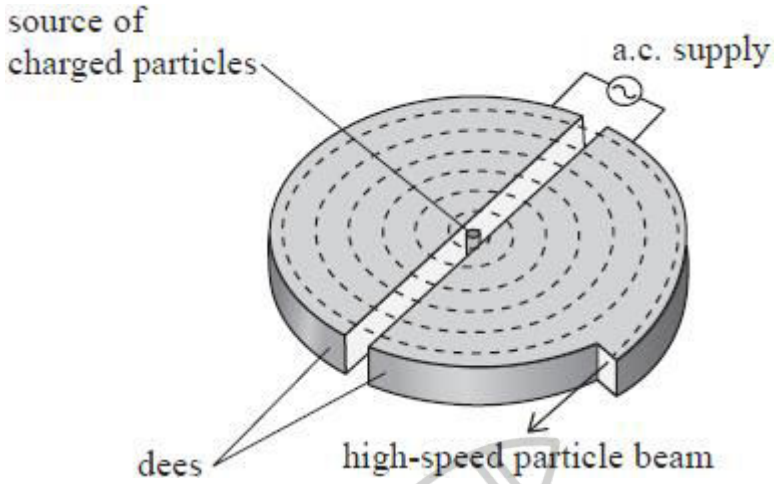
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K^- meson mass = ... kg

(b) The structure of a cyclotron is shown.



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(i) Explain how the cyclotron produces a beam of high energy particles.
You should refer to the alternating potential difference and the magnetic field.

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(ii) The first cyclotron was made by Lawrence in 1929. This cyclotron was able to produce a beam of protons with a maximum kinetic energy of 80 keV.
Calculate the magnetic flux density of the magnetic field used with this cyclotron.
diameter of cyclotron = 0.11 m

(4)

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Magnetic flux density = ...

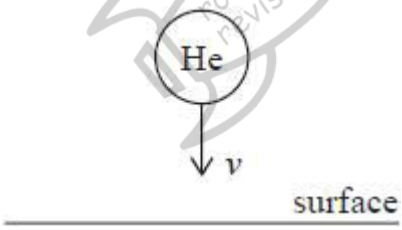
(Total for question = 13 marks)

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Q27.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

A helium atom of mass *m* and velocity *v* collides at right angles with a surface, as shown.



The atom bounces off the surface with the same speed but in the opposite direction. The collision takes time Δt.

Which of the following is the mean force acting on the atom during the collision?

☐ **A** $\frac{mv}{\Delta t}$

☐ **B** $\frac{2mv}{\Delta t}$

☐ **C** $\frac{-mv}{\Delta t}$

☐ **D** $\frac{-2mv}{\Delta t}$

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(Total for question = 1 mark)

Q28.

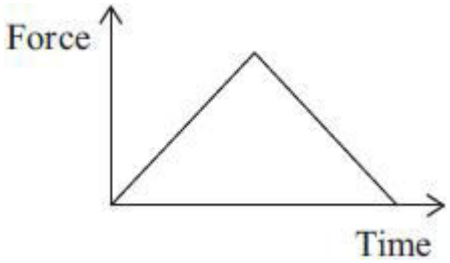
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Answer the question with a cross in the box you think is correct ☒). If you change your mind about an answer, put a line through the box ☒) and then mark your new answer with a cross (☒).

A ball collides with a wall and moves off in the opposite direction. The wall exerts a force on the ball during the collision.

The force-time graph for the collision is shown.



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Which of the following is given by the area under the graph?

- ☒ **A** acceleration of the ball
- ☒ **B** average force acting on the ball
- ☒ **C** change in momentum of the ball
- ☒ **D** distance travelled by the ball

(Total for question = 1 mark)

Q29.

Particle physicists use the principle of conservation of momentum when analysing particle interactions.

(a) State what is meant by the principle of conservation of momentum.

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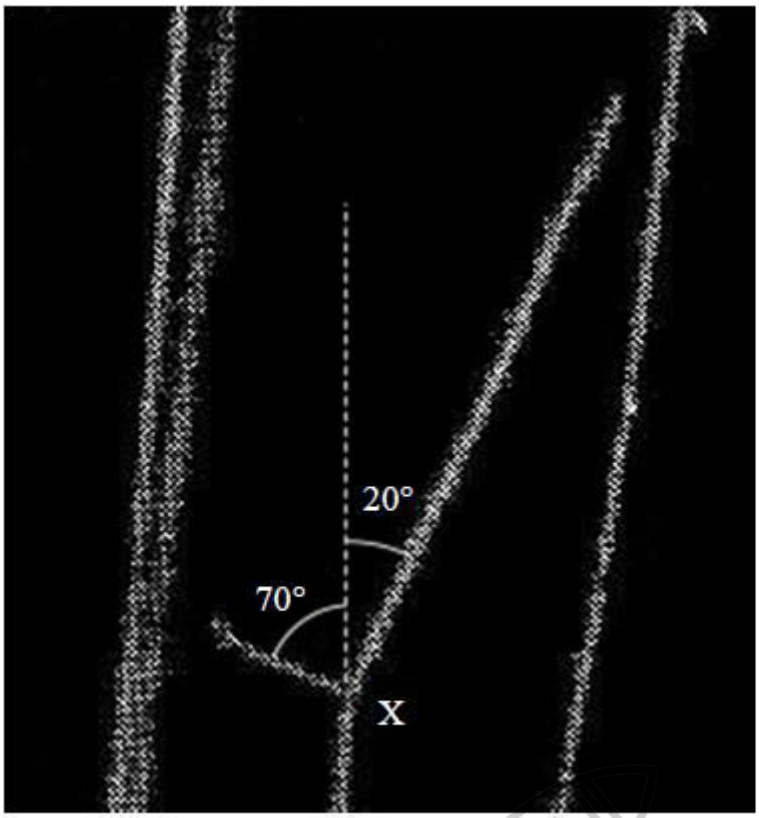
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(b) The cloud chamber photograph shows alpha particle tracks in helium.

An alpha particle collided with a helium nucleus at X. After the collision, the alpha particle moved at an angle of 20° to its initial direction, as shown.

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(Source: © Institute of Physics)

(i) Show that the speed of the helium nucleus immediately after the collision is about 5.6×10^6 m s^{-1}.
mass of helium nucleus = 6.64×10^{-27} kg
speed of alpha particle after collision = 1.55×10^7 m s^{-1}

(4)

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(ii) The kinetic energy of the alpha particle before the collision was 9.01×10^{-13} J.
Deduce whether the collision was elastic.

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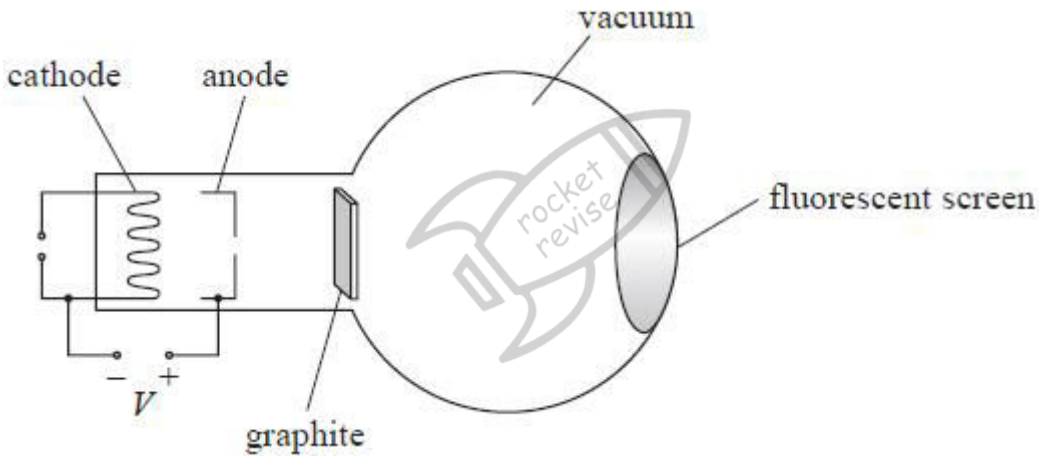
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(Total for question = 9 marks)

Q30.

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A beam of electrons was directed through a thin piece of graphite towards a fluorescent screen, using the apparatus shown.



A potential difference *V* was applied between the cathode and the anode to produce the beam of electrons.

A diffraction pattern was produced on the fluorescent screen.

From the diffraction pattern, the de Broglie wavelength of the electron beam was determined to be 2.55×10^{-11} m.

Calculate *V*.

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V = ...

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(Total for question = 4 marks)

Q31.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

An electron has a kinetic energy of 3.5×10^{-28} J.

Which of the following gives the momentum of the electron in kg m s^{-1}?

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☐ **A** $\dfrac{(3.5 \times 10^{-28})^2}{2 \times 9.11 \times 10^{-31}}$

☐ **B** $\dfrac{(9.11 \times 10^{-31})^2}{2 \times 3.5 \times 10^{-28}}$

☐ **C** $\sqrt{2 \times 1.6 \times 10^{-19} \times 3.5 \times 10^{-28}}$

☐ **D** $\sqrt{2 \times 9.11 \times 10^{-31} \times 3.5 \times 10^{-28}}$

(Total for question = 1 mark)

Q32.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

A moving trolley collides with a stationary trolley and they move off together.

Frictional forces are negligible and the collision is inelastic.

Which row of the table shows what happens to the total momentum and to the total kinetic energy in the collision?

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		Total momentum	Total kinetic energy
☐	A	conserved	conserved
☐	B	conserved	not conserved
☐	C	not conserved	conserved
☐	D	not conserved	not conserved

(Total for question = 1 mark)

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Q33.

The photograph shows a theme park attraction. Small electric cars are driven on a horizontal surface. The cars sometimes collide.



(Source: © Albin Raj Clicks/Alamy Stock Photo)

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(a) There is a thick rubber strip around the base of each car. Two cars collide and come to rest.

Explain why the rubber strip reduces the likelihood of injury to the drivers.
You should refer to change in momentum.

(3)

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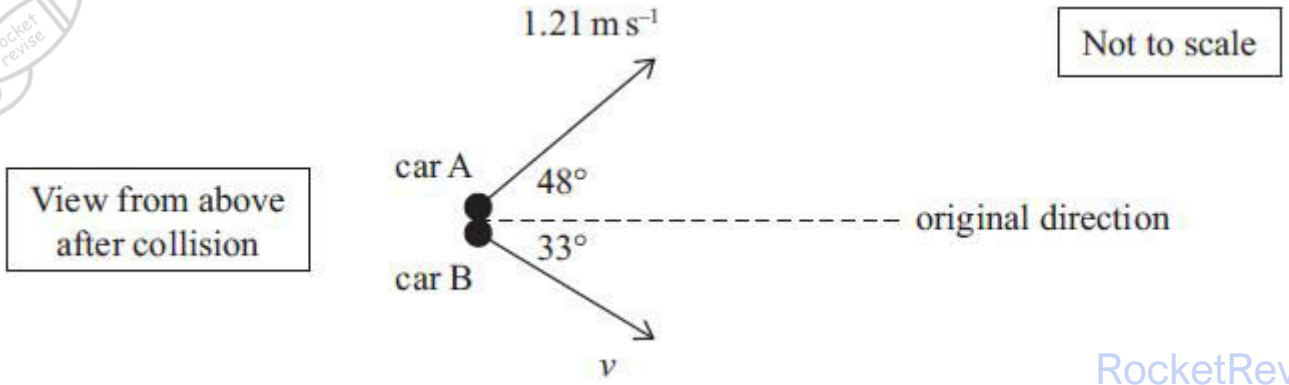
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(b) In another collision, car A collides with car B.

Car A is initially travelling at 2.19 m s^{-1} in a straight line. Car B is stationary.
After the collision, car A moves with a velocity of 1.21 m s^{-1} at an angle of 48° to its original direction.
Car B moves with a velocity *v* at an angle of 33° to the original direction of car A, as shown.



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Deduce whether the collision is elastic or inelastic.
mass of car A = 390 kg
mass of car B = 360 kg

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(Total for question = 9 marks)

Q34.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

In a nuclear fission reaction, a nucleus splits into a nucleus S, a nucleus T and several neutrons.

S and T have the same kinetic energy. T has twice the mass of S.

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Which of the following is the ratio $\frac{\text{momentum of T}}{\text{momentum of S}}$?

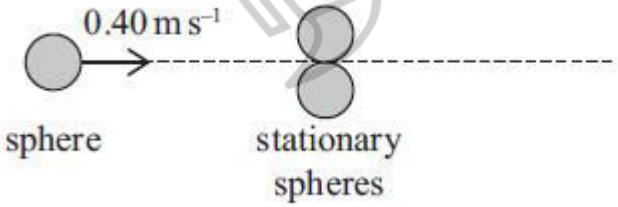
- ☐ **A** $\frac{1}{2}$
- ☐ **B** $\frac{1}{\sqrt{2}}$
- ☐ **C** $\sqrt{2}$
- ☐ **D** 2

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(Total for question = 1 mark)

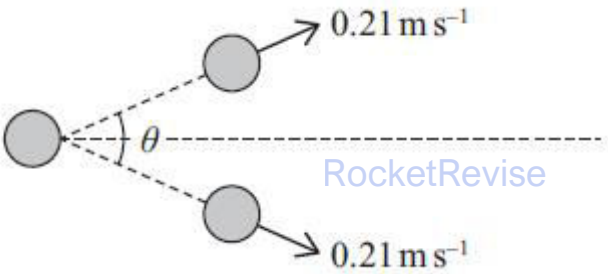
Q35.

A student investigated collisions, using three spheres. The student rolled a sphere of mass 35 g into two stationary spheres, each of mass 37 g, as shown.



During the collision, the rolling sphere was brought to rest and the two stationary spheres began to move apart at an angle θ, as shown below.

Not to scale



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Determine the angle θ.

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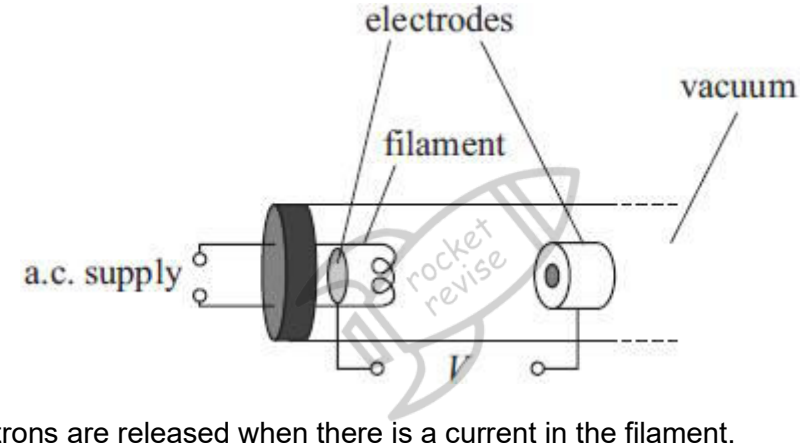
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θ = ...

(Total for question = 3 marks)

Q36.

A cathode ray tube can be used to demonstrate the properties of electrons. In the cathode ray tube, a beam of electrons is produced using the arrangement shown.



(a) Explain why electrons are released when there is a current in the filament.

(2)

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(b) In one cathode ray tube, the electrons in the beam had a de Broglie wavelength of 2.65×10^{-11} m.

Calculate the potential difference V required to accelerate these electrons between the electrodes.

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V = ...

(Total for question = 6 marks)

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